

FOAM-BORN WORMHOLE PAIRS IN FULL NUMERICAL RELATIVITY: MERGER, SCATTERING AND COLLAPSE AS A RACE BETWEEN COLLISION AND DECAY

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We evolve pairs of massive drainhole wormholes—phantom-scalar throats—in full numerical relativity and follow every outcome a pair can have. A lone throat is a fixed point with one unstable mode (e-fold time $\tau = 5.9$, in the González–Guzmán–Sarbach band) whose two branches, collapse and inflation, confirm the behaviour found for the massless throat; a declared kick picks the branch deterministically and accelerates it, forming a horizon at $t = 11$ for $\varepsilon = 10^{-2}$ against $t \simeq 61$ from truncation noise alone. Like-oriented throats repel—attraction requires flipping one scalar sign—with a force inverse-square in an effective separation one throat radius beyond the coordinate one. A head-on pair collides while both mouths are still open and makes a black hole, the trapped surface born enclosing both mouths and never either alone, which then loses mass to the phantom field it swallows ($M_{\text{MS}} 3.0 \rightarrow 2.2$) and rings without a chirp. The spiral merger instead stalls: no horizon exists anywhere on its measured record while the curvature wall that kills every grid rides the still-open throat, and a $16\times$ refinement ladder saturates without reaching the merger’s end. Capture ends between 50 and 70% of the circular momentum, and the loudest gravitational-wave channel of all is the horizonless fly-by, whose burst at $30 M_{\odot}$ would stand clear of the Advanced LIGO design floor at 10 Mpc.

I. INTRODUCTION

Wheeler’s spacetime foam is a sea of handles—wormholes that open and close on the Planck scale, both mouths in the same space [1–3]. More than one mechanism promotes them to macroscopic size: a Planck-scale Lorentzian wormhole caught in an inflating background is stretched with it and emerges macroscopic [4]; bubble and domain-wall cosmologies produce, without tuning, post-inflationary regions joined through wormholes [5, 6]; and a gas of relic wormholes has been run as a cosmology in its own right [7, 8]. What none of this work evolves is the aftermath: what a relic wormhole does once it is macroscopic and classical, and what two of them do to each other. That aftermath is this paper’s subject.

The classical theory fixes its two clocks. First, the simplest evolvable traversable wormhole—the Ellis–Bronnikov drainhole held open by a ghost scalar [9–12]—is unstable: perturbed one way the throat collapses to a black hole, the other way it inflates, on a timescale set by the throat radius [13, 14]. Every constituent destroys itself on its own schedule. Second, Geroch’s theorem forbids classical topology change [15, 16]: two wormholes cannot fuse into one, and none can be born. Birth is quantum [17], so a simulation takes over at $t = 0$ with the newborn state as its initial data—the stipulation every numerical-relativity binary makes, here with a physical story attached.

Together the two facts turn the standing objection to wormhole binaries into a selection rule. A binary has no astrophysical assembly channel—each throat dies faster

than capture could pair two—but a foam-born pair needs none: creation and collision are the same event, at a birth separation of order the objects themselves. The race is collision against decay, and both rates belong to the classical solution. The scalar’s sign adds a second rule at no cost: a phantom field repels like charges and attracts opposite ones, so like-signed pairs scatter while opposite-signed pairs fall together; and the sign of a lone throat’s perturbation selects collapse—a wormhole-to-primordial-black-hole channel beside the standard routes [18–20]—or inflation. A foam population thus fans out into four fates—merge, scatter, collapse, inflate—across the three-parameter space of birth separation, perturbation and relative sign, and that space is what a family of evolutions maps. The binary evolved here is the exterior approximation of a pair of handles (three asymptotic ends, the far sides passive); what merges is not two wormholes into one, which Geroch forbids, but two wormholes into one black hole with both throats inside it—the horizon does the work that topology change cannot.

The observable is not a chirp but a stochastic background from the merger epoch, redshifted to whichever band the mass scale puts it in [21–24]. The clean deliverable is therefore dimensionless and epoch-independent: the energy radiated per merger and its spectral shape in throat-mass units, which converts any bound on Ω_{GW} —including the nucleosynthesis cap that holds outside every detector band [25]—into a bound on the co-moving abundance of primordial wormhole mergers. Two caveats travel with the statement: the ghost scalar must exist at that epoch to hold the newborn throats open, and the merger’s scalar channel radiates negative energy whose cosmological consequence is genuinely open.

The paper is organised as follows. Sec. II sets out the model: the Einstein–phantom-scalar system, the drain-

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hole, the binary superposition with its sign rule, and the declared seeds. Sec. III gives the numerical setup and the run matrix. The results then follow the campaign from one throat to two: the lone wormhole—fixed point, unstable mode, branching and collapse (Sec. IV); two throats at rest and the force between them (Sec. V); the head-on merger to a black hole (Sec. VI); the spiral merger and its capture boundary (Sec. VII); the gravitational waves of every channel—collapse, head-on, spiral, fly-by—gathered in one chapter with the vacuum twin (Sec. VIII); the numerical controls and systematics (Sec. IX). Sec. X confronts the rescaled waveforms with the LIGO–Virgo–KAGRA data products—template banks, ringdown, burst searches—and Sec. XI collects the findings. Every evolution runs on GPUs with GRTeclyn [36], the AMReX-based [37] port of GRChombo [38], continuing the campaign of Refs. [39, 40]. Throughout, $G = c = 1$.

II. THE MODEL

A. Field equations

The matter is a single real, massless phantom scalar: gravity couples to *minus* the canonical stress tensor,

$$T_{ab} = -\left(\nabla_a \phi \nabla_b \phi - \frac{1}{2} g_{ab} \nabla_c \phi \nabla^c \phi\right), \quad V(\phi) \equiv 0, \quad (1)$$

so $\rho < 0$ where the field’s gradient dominates—the exotic stress that holds a throat open. With $V = 0$ this is exactly a ghost field. The sign flip lives entirely in the coupling to geometry: the scalar itself obeys the canonical wave equation $\square \phi = 0$, evolved first order in ϕ and $\Pi = (\partial_t \phi - \beta^i \partial_i \phi)/\alpha$.

B. Drainhole initial data

Each body is a massive Ellis–Bronnikov drainhole [9, 10]. In isotropic radius r about the throat centre, with

$$X = \frac{r - a^2/4r}{a}, \quad \Omega = 1 + \frac{a^2}{4r^2}, \quad u = \frac{m}{a} \left(\arctan X - \frac{\pi}{2} \right), \quad (2)$$

the exact static solution of Eq. (1) is

$$\alpha = e^u, \quad \gamma_{ij} = e^{-2u} \Omega^2 \delta_{ij}, \quad \phi = C \arctan X, \quad (3)$$

with $4\pi C^2 = a^2 + m^2$, $K_{ij} = 0$, $\Pi = 0$ and zero shift; in the evolved conformal variables $\chi = e^{2u} \Omega^{-2}$ and $h_{ij} = \delta_{ij}$. Both constraints hold identically at $t = 0$. Here a is the throat scale and m the ADM mass, carried by the *lapse* rather than the conformal factor, so χ at the throat stays in $[0.15, 0.25]$ for $m/a \leq 1$ and mass costs no grid resolution. The minimal surface sits at $r_t = \frac{1}{2}(m + \sqrt{m^2 + a^2})$ with areal radius $R_{\min} = e^{-u(r_t)} \sqrt{a^2 + m^2}$; the production throat $a = 2$, $m = 1$ has $R_{\min} = 3.8895$

at $r_t = 1.618$. The point $r \rightarrow 0$ is the far universe’s spatial infinity compactified ($\chi \sim r^4$). The mass matters: at $m = 0$ the throat is ultrastatic ($\alpha \equiv 1$, $M_{\text{ADM}} = 0$) and two of them do not attract at all.

C. The binary superposition

Two throats at separation d (one-body radii r_A , r_B) are superposed in excess form,

$$u = u_A + u_B, \quad \psi = 1 + (\sqrt{\Omega_A} - 1) + (\sqrt{\Omega_B} - 1), \quad (4)$$

with $\chi = e^{2u} \psi^{-4}$, $\alpha = e^u$ and $\phi = \phi_A + \sigma \phi_B$, $\sigma = \pm 1$; the constant tail of the arctan profiles is subtracted so $\phi \rightarrow 0$ at the boundary, which is free for a massless field. Momentum enters through one Bowen–York term [30] per throat,

$$\hat{A}_{ij} = \frac{3}{2r^2} [P_i n_j + P_j n_i - (\delta_{ij} - n_i n_j) P \cdot n], \quad (5)$$

converted by $\tilde{A}_{ij} = \chi^{3/2} \hat{A}_{ij}$, with $K = 0$. Since $\Pi = 0$ the matter momentum density vanishes and $\hat{A}_{ij}$ is flat-space divergence free, so the momentum constraint is satisfied *exactly* at $t = 0$; only the Hamiltonian constraint carries the superposition defect, $O(a^2/d^2) + O(m/d)$. The relative sign σ is the pair’s one discrete parameter: $\phi \rightarrow -\phi$ is exact for a single body, but a phantom mediator *repels* like scalar charges and *attracts* opposite ones, with $|F_\phi/F_{\text{grav}}| = (a^2 + m^2)/m^2 > 1$ always ($= 5$ for the production throat)—so like-signed pairs can never merge under their own gravity, and $\sigma = -1$ is the merger channel. A windowed one-body correction of the Helfer type [31] is implemented but off in production (Sec. IX A).

D. Declared seeds

The instability is probed with declared perturbations of the conformal factor only: a Gaussian shell on the minimal surface,

$$\psi \rightarrow \psi [1 + \varepsilon g(r)], \quad g = e^{-(r-r_t)^2/w^2}, \quad w = a/4, \quad (6)$$

which moves the throat’s areal radius by $\approx 2\varepsilon$ with the sign of ε , and a quadrupolar factor

$$\psi \rightarrow \psi [1 + \varepsilon_2 g(r) P_2(\cos \theta)] \quad (7)$$

about the z axis through the throat. K_{ij} and Π are untouched, so the momentum constraint stays exact. The spherical seed selects the branch; the quadrupolar seed has no $\ell = 0$ part—it gives a collapsing throat a quadrupole to shed—and with both on the cross term is $O(\varepsilon \varepsilon_2)$.

III. NUMERICAL SETUP

A. Evolution scheme

Evolutions use GRTeclyn: CCZ4 [32] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, $1 + \log$ slicing [33] started from the exact static lapse $\alpha = e^u$ of Eq. (3), a Gamma-driver shift [34, 35] with $\eta = 1$ started from zero, fourth-order stencils, RK4, and Kreiss–Oliger dissipation $\sigma = 0.1$. The time step is $\Delta t = 0.02 \Delta x$, an order of magnitude below the usual CCZ4 Courant factor; the bracket in Δt that makes it necessary is a control (Sec. IX A).

B. Grid, boundaries and regularisation

The domain is a cube of side $L = 64$ with the system at its centre; the base grid is 128^3 ($\Delta x = 0.5$). Refinement is by moving nested boxes centred on the tracked throats, shrinking by a factor of two per level (half-width 4 per throat at the production finest level) and re-drawn every 16 steps. Production runs use three refinement levels (finest $\Delta x = 1/16$); resolution arms raise this to levels 4–7 where stated, and one control doubles the box to $L = 128$. Outer boundaries are Sommerfeld, backed by an absorbing sponge over $r = 24$ –32 that ramps extra dissipation up to $\sigma_{\text{sp}} = 4$ as the fourth power of the depth into the layer; everything dynamical sits inside $r = 24$.

Each throat carries a compactified second infinity at its own centre, where $\chi \sim r^4$, so the evolved variables are floored: $\chi \geq 10^{-8}$ and $\alpha \geq 10^{-10}$ (a variant keeps χ unfloored down to 10^{-20} and applies the 10^{-8} floor only where $1/\chi$ is evaluated). Two further devices act only inside regions that are already trapped: an exponential damping of the scalar where the lapse has collapsed, and a smooth interior freeze that multiplies every right-hand side by $1 - W(r)$, with W a C^2 quintic step between an inner radius r_{full} and an outer r_{start} . All three are numerical devices, not physics; whether any of them is load-bearing is itself measured by twin runs (Secs. VIB and IX A).

C. Diagnostics

L^2 norms of the Hamiltonian and momentum constraints over the box are written continuously. Both throat centres are tracked, and about each centre the areal radius $R(r)$ is scanned on coordinate spheres; its minimum is the throat radius. On the same spheres the two null expansions $\theta_{\pm}$ are computed with the outward direction fixed by *increasing areal radius*: a marginally outer trapped surface is an outermost zero of θ_+ with $\theta_- < 0$ (crossings at extrema of $R(r)$ are rejected), while a dynamic wormhole throat is marginally *anti-trapped* [29]. Masses of such surfaces are Misner–Sharp [28], $2M/R = 1 + \frac{1}{4}R^2\theta_+\theta_-$; the scan uses coordinate spheres rather than a deformable finder, and

TABLE I. The run matrix: 111 evolutions, grouped. Every group varies one knob against a named reference; per-run lines live in the shipped registry. Body parameters are $a = 2$, $m = 1$ ($R_{\text{min}} = 3.89$), $\sigma = -1$ and $d = 12$ unless stated.

group	knob(s)	cells	Sec.
lone throat	levels 2–4; χ floor; Δt ; $L=128$	9	IV
spherical kicks	$\varepsilon = \pm 10^{-3} \dots \pm 10^{-1}$; level 4	8	IV C
quadrupolar kicks	$\varepsilon_2 = 0.01, 0.05$; levels 3–4	4	VIII A
two throats at rest	σ ; $a = 1$ –3; $d = 12$ –18	9	V
placement probes	$d = 6$ –48, one step each	18	V C
head-on, $d = 8$	fill; levels 3/5; down-step	8	VI
half-mass, $m = 0.5$	pair at $d = 6$; lone control	2	VIC
spiral, $p = 0.12$	levels; N ; gauge; floor; fill	29	VII
capture scan	$p = 0.20$ –0.45	4	VII A
Helper twins	correction; window; levels	6	IX A
vacuum BBH	same d , p , masses; no scalar	2	VIII E
shakedown (archived)	gauge, dissipation, tagging	12	III A

quoted masses carry that shape systematic. Radiation is read from Ψ_4 interpolated onto spheres at $R = 10, 14, 18$ (and 22 on the single-throat wave arms), decomposed in spin-weight -2 harmonics; all amplitudes are quoted as $r\Psi_4$. The in-code stream is cross-checked against an independent plotfile pipeline at the same radii. A per-cell autopsy reports the location and one-step history of any floating-point failure.

D. The campaign

The campaign is one-knob: every run differs from a named reference in a single setting, and every run has one line—its difference, its outcome, its caveats—in a registry shipped with the results pack (Sec. XII). Table I groups the 111 evolutions; the verdicts are the business of Secs. IV–IX. An archived shakedown stage (12 of the 111) fixed the gauge, dissipation and tagging choices of Secs. III A–III B and is not used for physics.

IV. THE SINGLE WORMHOLE

A. The static throat as a fixed point

Evolved with no declared perturbation, the exact drainhole holds its areal throat radius $R_* = 3.8895$ to better than 0.1% for 35 time units at level 3 and to 1% until $t \simeq 44$; at level 4 the radius is still exact to four digits at $t = 30$. The static throat is a fixed point of the discrete system as well as of the continuum one. What ends the quiet phase is not a failure of the data: the departure time moves *later* by 5.6–8.7 units per halving of the cell, the signature of a truncation-noise seed of fixed growth rate being handed a smaller initial amplitude. Level 2 never reaches the branch at all—it dies off

an origin failure at $t = 24.2$ with the throat still exact to 0.12%.

B. The unstable mode

The departure is a single exponential mode (Fig. 1): plateau fits give e-fold times $\tau = 5.88$ (level 3) and 5.26 (level 4)—the same mode to 9%—at $T = \tau \alpha_{\text{th}}/R_* = 0.87$ and 0.78 against the González–Guzmán–Sarbach band 0.68–0.76 for the massless limit [14]. The instability is the throat’s; only its seed is the grid’s, and the seed’s *sign* flips with resolution: level 3 collapses, level 4 inflates. This confirms, from exact massive-drainhole data, the unstable behaviour and the two-branch structure found for the massless Ellis–Bronnikov throat in Ref. [39], where ambient noise drove inflation and a compressive perturbation drove collapse: the branches are the wormhole’s, not an artefact of either code.

C. Branching: collapse versus inflation

A declared kick removes the grid’s vote (Fig. 2). At $\varepsilon = \pm 10^{-2}$ and $\pm 10^{-3}$ the fate is always opposite to the push, each $\pm$ pair crosses itself at the ε -independent $t_{\times} \simeq 13.0$, and a tenfold smaller kick arrives 8–12 units later—linear response riding the same unstable mode. The kick also sets the clock: the collapse branch forms its horizon at $t = 11$ for $\varepsilon = +10^{-2}$ and $t = 25$ for $+10^{-3}$, against $t \simeq 61$ for the unkicked throat left to truncation noise—a driven branching an order of magnitude faster than the undeclared one. The dial has a ceiling: $\varepsilon = \pm 0.1$ is not a perturbation—both signs collapse and die ($t = 15.2/14.1$) with a Hamiltonian violation 3–5 $\times$ the 1% arms’ and growing—so the usable amplitude ends at 10^{-2} .

D. The collapse endpoint

The collapse branch ends in a black hole that eats its own support: behind the marginally trapped surface the areal radius falls $3.23 \rightarrow 2.37$ and the Misner–Sharp mass $1.62 \rightarrow 1.23$ over 40 units—phantom infall *removes* mass—with the origin lapse at 0.016 and no bounce by $t = 100$. (The massless throat of Ref. [39] bounced at $t \approx 4M$ and shed its horizon by $18.5M$; the massive drainhole shows no bounce in 40 units.) The kicked $\varepsilon = +10^{-2}$ arm runs clean to $t = 100$ behind its horizon. The inflation branch forms no horizon at all: a growing anti-trapped shell, the throat +122% by $t = 100$, the compactified inner sheet deforming. Mass sets the stability margin from below: the lone $m = 0.5$ throat tracks its exact solution only to $t \approx 17$ at the 0.1% level (the $m = 1$ twin holds to $t \approx 35$) and dies at its origin at $t = 26.3$ with no horizon—the lighter throat is the less stable one.

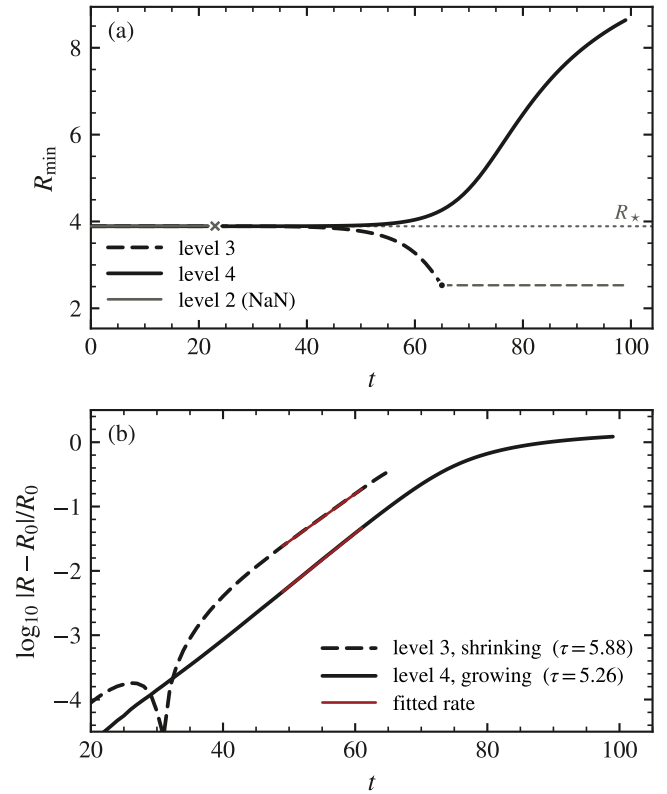


FIG. 1. The undeclared seed: one exact throat, two resolutions, two fates. (a) Minimal areal radius of the lone drainhole evolved from exact static data at refinement levels 2–4. The exact solution is a fixed point (R_* , dotted); each level holds it until its own truncation error excites the unstable mode, and the fate flips with the grid: level 3 (dashed) collapses, level 4 (solid) inflates. Level 2 (grey) dies of an origin failure at $t = 24.2$ (cross) with the throat still exact to 0.12%—too coarse to reach the branch at all. Dot and muted line as in Fig. 2: the scan loses the level-3 throat at $t = 65$ while the run continues to $t = 100$ behind a marginally trapped surface (areal radius $3.23 \rightarrow 2.37$). (b) The deviation grows exponentially on both grids: plateau fits over $t = 49$ – 61 give e-fold times $\tau = 5.88$ (level 3) and 5.26 (level 4)—the same unstable mode to 9%, $T = \tau \alpha_{\text{th}}/R_* = 0.87$ and 0.78 against the González–Guzmán–Sarbach band 0.68–0.76 [14]—with opposite sign. The fate is the truncation seed’s; the mode is the throat’s.

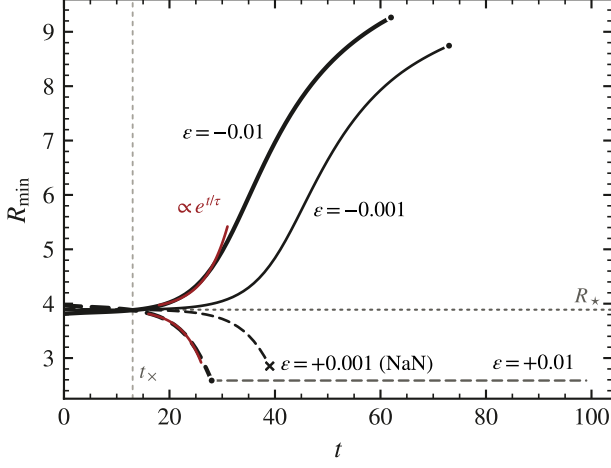


FIG. 2. Declared-seed branching of the lone throat (level 3). Areal throat radius of the drainhole ($a = 2$, $m = 1$; $R_* = 3.8895$, dotted) under the kick of Eq. (6): $\varepsilon = \pm 10^{-2}$ (heavy), $\pm 10^{-3}$ (light); dashed = outward ($\varepsilon > 0$). The fate is opposite to the push, and each pair crosses itself at the ε -independent $t_\times \simeq 13.0$ —linear response. Burgundy: exponential fits $|R - R_*| \propto e^{t/\tau}$, $\tau = 4.5$ ($t = 18$ – 31) and 4.0 ($t = 16$ – 26)—steeper than the unkicked mode’s 5.88 (Fig. 1) while the kick’s transient decays. Dots: the scan loses the throat (the inflating runs continue to $t = 100$, still growing); cross: death at $t = 40.1$. The $\varepsilon = +10^{-2}$ arm holds a horizon from $t = 11$ (areal radius $3.88 \rightarrow 2.34 \rightarrow 2.57$) and runs clean to $t = 100$; the muted line carries its last reading. A tenfold smaller kick arrives 8 – 12 units later.

V. TWO THROATS AT REST

A. The sign rule

Two identical throats released from rest push apart; reversing one throat's scalar field turns the push into a pull (Fig. 3). That one sign is the only gravity-driven route to a merger in this model, and every merging binary in this paper carries it. The measured pull-to-push ratio at $a = 2$ is 1.51 ± 0.03 against the predicted $3/2$, and the force follows the throat *width*—the scalar charge—not the mass, which is equal in all arms.

B. The force law

Four like-signed pairs, byte-identical but for the gap, give displacements at a common $t = 11.5$ of $0.4696/0.3699/0.2980/0.2438$ for $d = 12/14/16/18$. Pure $1/d^2$ is excluded— $F d^2$ rises monotonically by 15.4% across the ladder—but $F \propto (d + \delta)^{-2}$ with $\delta \simeq 3.5$ fits every rung pair, and it was a blind prediction: δ fitted on $d = 12$ – 16 alone predicts 0.243 at $d = 18$ where inverse-square predicts 0.209, and the measurement is 0.2438. Since δ is comparable to the measured physical throat radius (4.29 at $a = 2$), the force is inverse-square in an *effective* separation exceeding the coordinate one by about a throat radius—the deviation lives in the distance label, not in the law. Across widths the push grows at every rung ($0.147/0.283/0.416/0.615$ for $a = 1/1.5/2/3$) with a measured exponent ≈ 1.36 and falling; the point-charge a^2 scaling is not confirmed.

C. Placement systematics

The superposition itself is on the ruler: freshly placed pairs read +1.8% ($d = 48$) to +20.5% ($d = 6$) wider than the isolated throat, falling as $d^{-1.16}$ (Fig. 4), so every same- d comparison in this paper is made against that curve rather than against the lone throat. Read that way, the released head-on pair's mouths are genuinely squeezed, -0.6% at $t = 7$ growing to -4.0% by $t = 13$ —an interaction effect, not placement, since the lone throat is exact until $t = 35$.

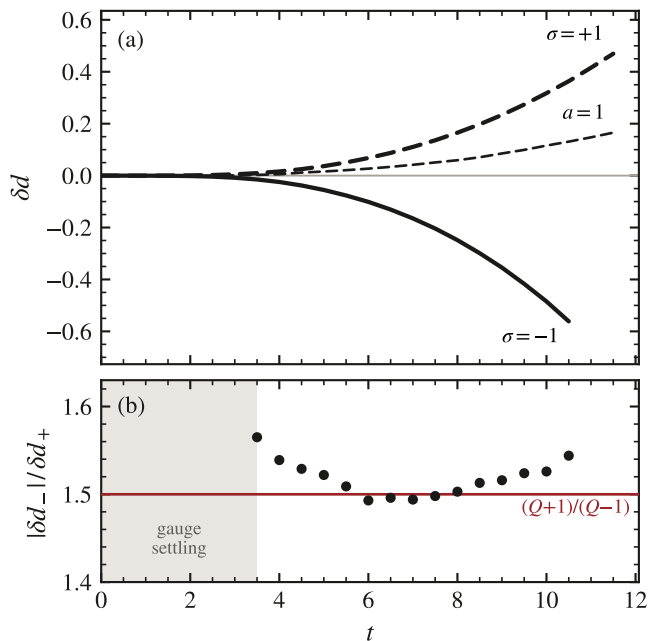


FIG. 3. Attraction and repulsion switched by one sign ($d = 12$, released from rest). (a) Separation change of three pairs one knob apart: the like-oriented pair ($\sigma = +1$, dashed) pushes apart; flipping one field ($\sigma = -1$, solid) turns the push into a pull; the narrow like pair ($a = 1$, light) opens $2.9\times$ more slowly—the force follows the throat width (the scalar charge), not the mass, which is equal in all three. Separations are inverse- χ -weighted centroids of the midplane χ pits, sub-cell where the mouth tracker's 0.03 grid quantum would swamp the early motion. (b) The flipped pair's infall over the like pair's escape: with $Q \equiv |F_\phi/F_{\text{grav}}| = 5$ (Sec. II C), pull and push are $(Q+1)$ and $(Q-1)$, predicting $3/2$ (burgundy); measured 1.52 ± 0.02 over $t = 3.5$ – 10.5 . Coordinates under-read the motion while the gauge settles (band), but that systematic is common to both arms and cancels in the ratio: the two runs differ in exactly one parameter.

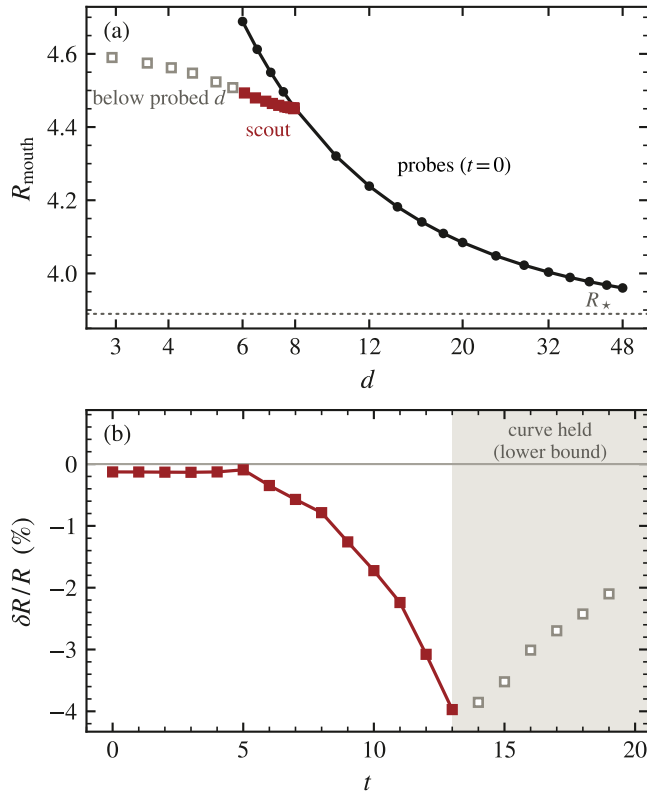


FIG. 4. Placement is on the ruler; the squeeze is not. (a) Per-mouth areal radius of two exact drainhole throats placed at rest, read at $t = 0$ by one-step probes at separations $d = 6$ –48 (ink): the pair reads wider than the isolated throat (R_* , dotted), the excess falling as $d^{-1.16}$ —the neighbour’s field on the ruler, not a dynamical widening. Burgundy squares: the head-on scout’s mouths (the $\sigma = -1$ pair released from rest at $d = 8$, Sec. VI) at the separation it had reached; open squares, here and in (b): the scout closer than the closest probe. (b) The scout against the curve at its own separation, $\delta R/R = R_{\text{scout}}/R_{\text{place}} - 1$. The -0.13% plateau to $t = 5$ is the mouth tracker’s grid snap (d reads 7.94 for a true 8) and calibrates the systematic; the throats’ own squeeze then grows to -4.0% by $t = 13$ ($d = 6.1$). In the band the curve is held at its last point, so the magnitude is a lower bound—the apparent recovery is the ruler’s, not the throats’. A boost can only raise a coordinate-sphere reading, so a reading below the curve is a real squeeze, if anything underestimated.

VI. THE HEAD-ON MERGER

A. Contact and the common horizon

Released from rest at $d = 8$ with opposite scalar signs, the pair closes to contact at $t \approx 20$ while both mouths are still open wormholes, and a common marginally outer-trapped surface encloses them from $t = 22$: the offline corrected scan finds it at $r = 3.17$, areal radius $R = 5.56$, Misner–Sharp mass 2.99, growing to $R = 5.71$ by $t = 24$ (Fig. 5). The per-mouth scans never find a surface at either mouth at any time—the horizon is born common or not at all. The formation is robust: the frozen-interior ensemble, including both signs of the seed ε , finds the same surface at $t = 22.0$ with its radius within half a percent across arms.

B. Through the wall to a black hole

Five units after the horizon forms, the level-3 scout dies at $t = 26.91$ with the classic anatomy— χ floored at the midpoint, $\max|K|$ running away to 18, a one-step overflow—but the failure is the grid’s, not the space-time’s: at $t = 27.05$, where the scout died, the level-5 arm reads a smooth, monotone midpoint (lapse 1.6×10^{-2} , $\chi = 1.2 \times 10^{-6}$, $\max|K| = 0.11$) and runs to $t = 100$ with the constraint norms *falling* the whole way. The remnant is a trapped ball with no throat left, losing mass to the phantom field it swallows ($M_{\text{MS}} 2.99 \rightarrow 2.17$, Fig. 5). The exterior is grid- and device-independent: the down-step arm lands on the level-5 constraints to 0.08%/1.5% and on its wave to 0.05–0.33% of peak over $t = 45$ –100, and the fill twins agree with the no-fill arms at the 0.01%-of-peak level.

C. The half-mass pair

Halving the throat mass does not avoid the horizon; it moves everything earlier. At $m = 0.5$, $d = 6$, the lapse falls from 0.44 to 0.036 over $t = 7.5$ –13, the tracker merges the mouths at $t = 11.1$, the offline scan confirms a common MOTS at $t = 12$ (areal $R = 3.83$, $M_{\text{MS}} = 2.16$), and the run dies at $t = 14.0$. The death belongs to the pair: the lone $m = 0.5$ control is exact to 0.02% through $t = 14$ and only fails, with no horizon of its own, at $t = 26.3$.

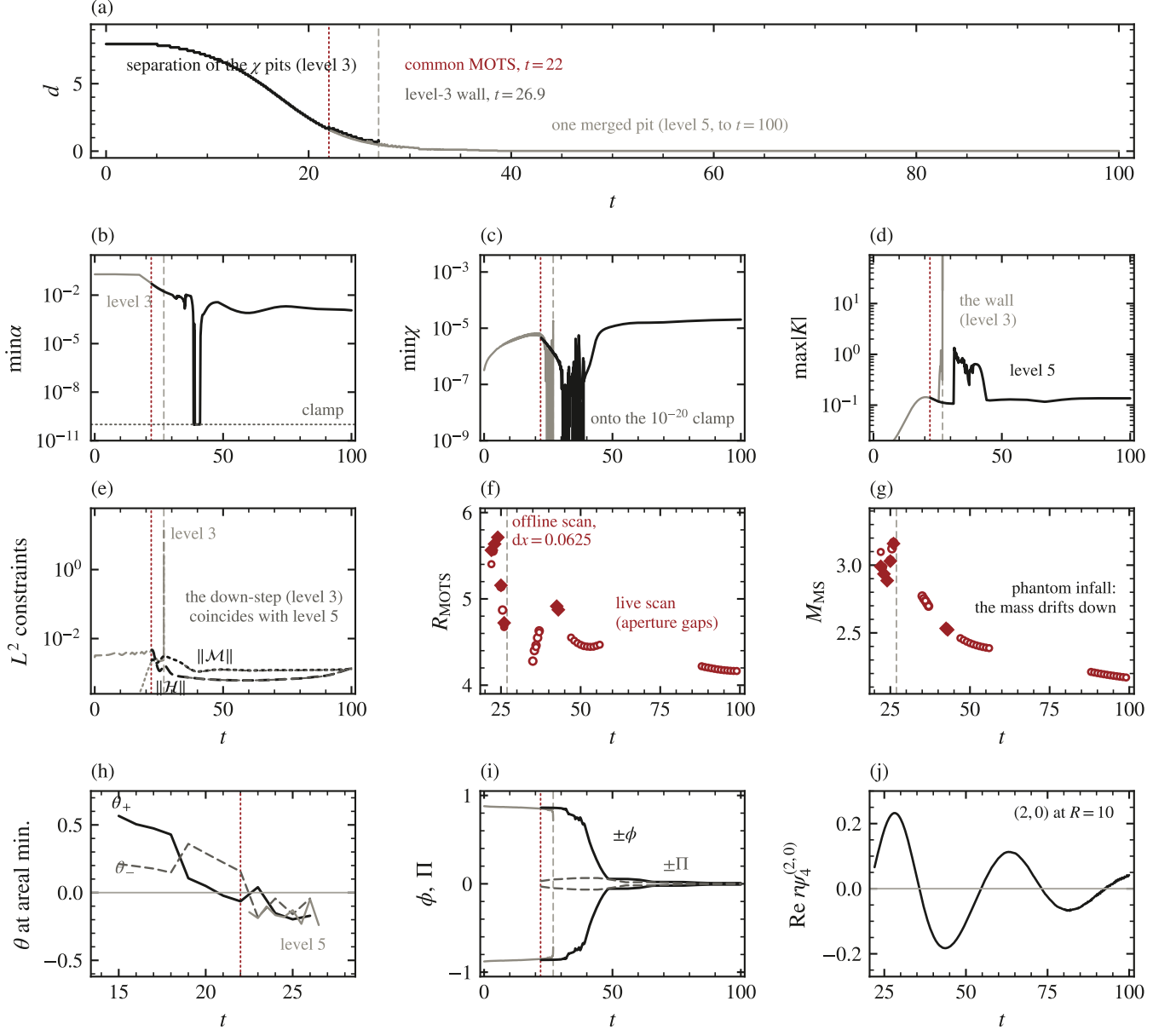


FIG. 5. The head-on merger makes a black hole ($d = 8$ from rest, opposite scalar signs); every horizon point is the orientation-corrected scan. (a) Separation of the χ pits: level 3 carries the formation from $d = 7.94$ (the tracker's grid snap on a true 8); the common MOTS forms at $t = 22$ (burgundy rule in every time panel), the scout dies at the level-3 wall at $t = 26.91$ (dashed rule), and the level-5 arm, restarted at $t = 22$ with every interior device off, carries the merged pit to $t = 100$ with no NaN. (b) The minimum lapse (level 5, ink) rides the 10^{-10} clamp over $t = 38.5\text{--}41.0$ and settles at 1.1×10^{-3} . (c) Minimum χ : excursions onto the 10^{-20} clamp leave the frame; level 5 returns and settles at 2.0×10^{-5} . (d) The wall is resolution: $\max|K|$ blows up to 18 at the level-3 death; level 5 never exceeds 1.34 and ends at 0.14. (e) The scout's $\|\mathcal{H}\|$ explodes at the wall (13.6); the level-5 norms fall $4.6 \times 10^{-3} \rightarrow 1.4 \times 10^{-3}$; the down-step (level 3 again, from the $t = 35$ checkpoint) lands on them to median 0.08%/1.5%. (f) The horizon: filled diamonds, offline scans at $dx = 0.0625$ (formation at $t = 22$, $r = 3.17$, $R = 5.56$; growth to 5.71 by $t = 24$; then 4.72, and 4.92/4.87 at $t = 42.5/43$); open circles, the live scan on three arms. Its gaps are aperture, not horizon loss—the shells reach $0.5d + 2.3$, so once the pits merge the surface at $r \simeq 2.7\text{--}3.3$ lies outside while every shell within reports trapped; where two arms see it at once they agree ($t = 36$: $R = 4.461$ against 4.441). (g) The Misner–Sharp mass drifts down, 2.99 at formation to 2.17 at $t = 99$ (-0.0066 per unit): phantom infall *removes* mass. (h) Expansions at the areal minimum: θ_+ crosses zero between $t = 20$ and 21, both negative by $t = 24$; the per-mouth scans never find an own MOTS—the horizon is born common. (i) The field's extrema fall $\pm 0.86 \rightarrow \pm 0.008$ with $|\Pi| \leq 0.07$: the supporting scalar is swallowed and radiated, leaving vacuum around a black hole. (j) The remnant's $(2,0)$ mode at $R = 10$ rings with period $\sim 31\text{--}38$, $\times 0.6\text{--}0.8$ per half-swing, as the mass in (g) declines.

VII. THE SPIRAL MERGER

A. Capture and scattering

Dialling the tangential momentum at $d = 12$ splits the outcomes three ways: $p = 0.12$ and 0.15 fuse ($p = 0.15$ reproduces $p = 0.12$'s endgame signature—a plateau at pit separation 0.816 against 0.815, then the dive with the core lapse rising); $p = 0.20$ and 0.25 are captured into bound orbits; $p = 0.35$ and 0.45 escape. The capture boundary therefore sits between $p = 0.25$ and 0.35 —50–70% of the circular value—and the fusing boundary between 0.15 and 0.20 . The $p = 0.45$ fly-by is its own result: a bound pass at closest approach 3.95 after which *no horizon ever forms*—the corrected scan reads zero surfaces on the whole record—and both mouths swell $\times 7.8$ in areal radius, that expansion, not any wall, driving the late constraint growth ($\text{corr}(\log R, \log \|\mathcal{H}\|) = 0.92$).

B. Inspiral and collapse

The production spiral ($p = 0.12$, $L = 128$) inspirals from $d = 11.9$ to a single merged pit and then collapses at the centre while the throat stays open (Fig. 6). The merger's constraint transient at $t = 41.2$ is resolved, not resolution—the level-3 leg reproduces the level-5 norms to 3.5%/0.7% over the shared window—and everything after it is a clean gauge-and-curvature collapse: the lapse to 2.0×10^{-3} , χ to its floor at $t = 58.4$, $\max |K|$ up $25\times$ on a wall that lands on the throat. Unlike the head-on, the corrected scans find *no* marginally outer-trapped surface at any of $t = 55$ – 60 , at either resolution, while the throat's areal radius shrinks only 5.7%: at its death this remnant is still a wormhole.

C. The wall

No grid gets the spiral through its merger. Restarting the $p = 0.12$ run from the same $t = 50$ checkpoint at `max_level` 3–7, every arm dies on the same h_{11} NaN, and the gains saturate: +1.03, +2.51, +0.53, +0.06 units per doubling (Fig. 7). Sixteen times the resolution buys 4.13 units and never reaches the merger's end, and a halved time step buys +0.36 on its own level-5 twin—most of the level $5 \rightarrow 6$ spatial gain—so the top of the ladder is saturation evidence, not a converged limit. The wall is insensitive to the matter device (the damped family dies on the same clock), to starting refined (`max_level` 5 from $t = 0$ dies at $t = 52.07$, indistinguishable from the restarted rung), and it moves only modestly when the whole problem is re-based: the production chain—base grid doubled, restarted three times earlier at $t = 36$ —buys 4.2 units past the entire ladder and still dies, at $t = 60.445$. The interior freeze then carries the exterior to $t = 100$; it is a

numerical device, engaged only after the burst is causally clear of the extraction spheres (Sec. VIII C).

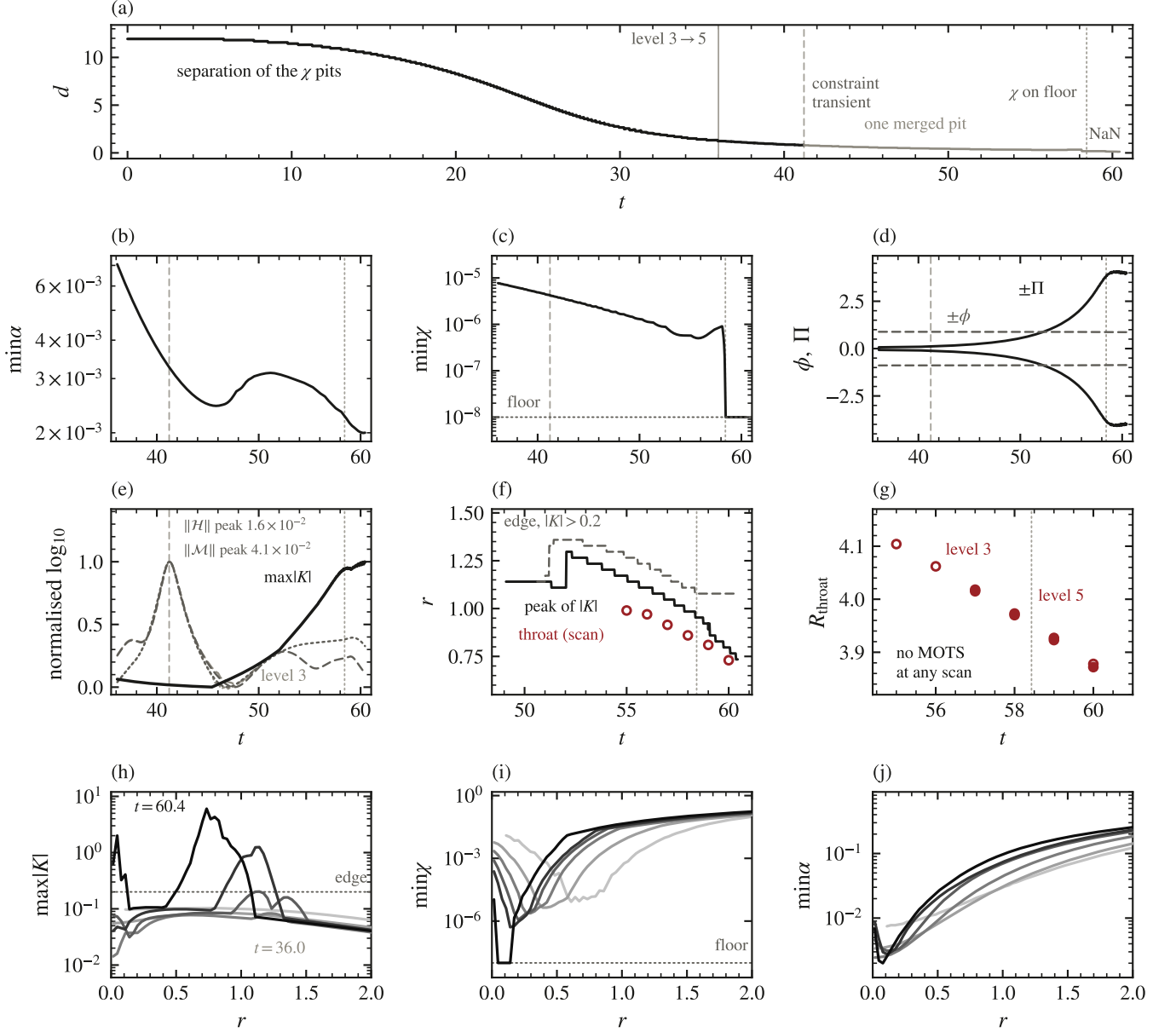


FIG. 6. The production spiral's collapse ($p = 0.12$, $d = 12$): the level-5 window from the restart at $t = 36$ to the death by NaN at $t = 60.44$, all of it unfrozen physics—the interior fill plays no part in any panel. (a) Coordinate separation of the two χ pits over the whole history: the level-3 leg carries the inspiral from $d = 11.9$ to the handover at $t = 36$ (solid rule); the constraint transient at $t = 41.2$ (dashed rule here and in every time panel) marks the merger, after which the tracker follows a single merged pit (pale); the record ends at the NaN. (b) The minimum lapse bottoms at 2.0×10^{-3} and never reaches zero. (c) The minimum conformal factor decays exponentially and sits on the code's 10^{-8} floor from $t = 58.43$ (dotted rule in every time panel), at $r = 0.078$. (d) The scalar field's extrema hold ± 0.89 throughout while its momentum saturates at $\Pi = \pm 4.1$: the profile survives, the momentum diverges. (e) Growth compared, each curve $\log_{10}$ -scaled onto $[0, 1]$: the constraint norms peak at the merger and relax below where they started, while $\max |K|$ only begins its climb near $t = 49$ —after $t = 40$, $\text{corr}(\log \max |K|, \log \|\mathcal{H}\|) = -0.35$ and $\text{corr}(\log \max |K|, \log \|\mathcal{M}\|) = +0.04$, so the death is not a constraint blow-up. The pale curves are the level-3 leg's norms through the same transforms: they lie on the level-5 curves to a median 3.5% ($\mathcal{H}$) and 0.7% ($\mathcal{M}$) over the shared window $t = 36$ – 50 —the transient is resolved, not resolution. (f) Where the radius of the $|K|$ spike (steps; drawn from $t = 49.1$, before which the profile is flat and an argmax is noise) and the disturbance's outer edge $|K| > 0.2$ (dashed), widest at $r = 1.36$ ($t = 51.4$) and contracting after. Burgundy circles: the throat's coordinate radius from the orientation-corrected marginal-surface scans—the spike rides just outside the throat and lands on it by the death. (g) The same scans' throat areal radius: R shrinks $4.10 \rightarrow 3.87$ (-5.7%) over $t = 55$ – 60 , levels 3 and 5 agreeing to 0.2%, and *no* marginally outer-trapped surface exists at any scan—the collapse runs to the end with no horizon found. (The in-code θ_+ columns are not drawn: their naive $+r$ orientation manufactures trapped verdicts near a throat, the artefact behind a withdrawn early horizon claim.) (h)–(j) The radial anatomy at six equally spaced times, light ($t = 36.1$) to dark ($t = 60.4$): the per-shell $\max |K|$ grows $0.08 \rightarrow 6.1$ on a spike ~ 0.25 wide migrating inward (h), while the χ pit (i) and the lapse pit (j) stay at $r \lesssim 0.15$. The gauge collapse and the curvature wall are different places: the pit at the centre, the wall on the throat.

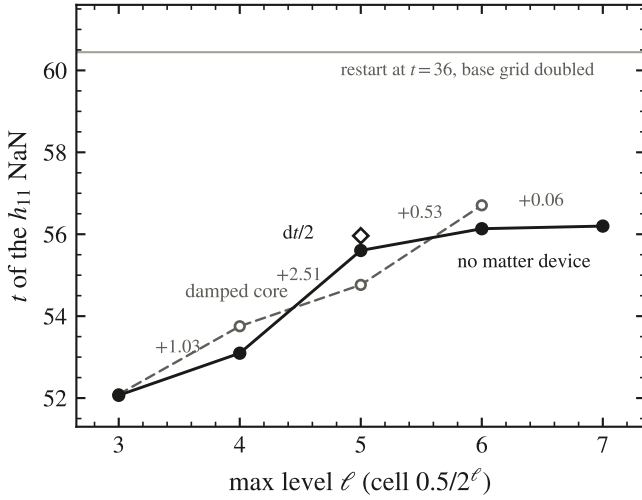


FIG. 7. The wall does not move: death time of the $p = 0.12$, $d = 12$ spiral against refinement, every arm restarted from the same $t = 50$ checkpoint, every death an h_{11} NaN, no horizon found at any rung. Filled: plain evolution, with the per-doubling gains marked—+1.03, +2.51, +0.53, +0.06; the last doubling of a $16\times$ ladder buys 0.06 units. Open diamond: the level-5 arm re-run at half the time step (+0.36, most of the level $5 \rightarrow 6$ spatial gain). Open circles: the lapse-keyed core-damping family—a different treatment, listed separately, dying on the same clock. The faint rule is the production chain (base grid doubled, restart at $t = 36$): 4.2 units past the whole ladder, and still short of the merger’s end.

VIII. GRAVITATIONAL WAVES

Every channel radiates, and the loudest is the one that never makes a horizon. At the innermost sphere the peak $|r\Psi_4|$ of the dominant mode reads: fly-by 4.1×10^{-2} , spiral 2.9×10^{-2} , head-on 2.3×10^{-2} , vacuum twin 1.1×10^{-2} , collapsing throat 2.9×10^{-3} (Fig. 8)—the scattering pass outradiates every merger, by $1.4\times$ over the spiral and $3.8\times$ over the equal-parameter vacuum binary. Each record propagates at c within the output cadence and falls as $1/r$ across its spheres, and rescaled to a detector the ordering survives: the fly-by’s strain peaks at $1.7 \times 10^{-20} \text{ Hz}^{-1/2}$, the mergers within a factor 2.4 of one another an order below it, the lone throat an order below them (Fig. 9).

A. The collapsing throat

A lone throat radiates only if its collapse is not spherical. The spherically kicked arm emits nothing above the $\ell = 2$ numerical floor—by symmetry—while the same kick dressed with an $\varepsilon_2 = 0.05$ quadrupole produces a burst of peak $|r\Psi_4| = 2.9 \times 10^{-3}$, $35\times$ the no-quadrupole control, carrying $E_{\text{rad}} = 2.3 \times 10^{-6} M$. The burst arrives in radius order at $0.8\text{--}1.0c$, carries the same $r\Psi_4$ to four spheres within 7.2%, and scales by 4.85 for a $5\times$ kick—

linear response, so the source is the seeded quadrupole and not the collapse itself. What is left rings at period $20.6 M$, 21% short of the Schwarzschild fundamental for the horizon mass; the decay time is not measurable on this record.

B. The head-on ringdown

The head-on’s signal is a bell with no chirp: the $(2, 0)$ breathing mode at $R = 10$ swings through +0.23, -0.18 , +0.11, -0.07 at $t = 28.2, 43.6, 63.0, 81.7$ —a period of $\sim 31\text{--}38$, lengthening, with the amplitude falling $\times 0.6\text{--}0.8$ per half-swing—and the same bell arrives ~ 4 units later per 4 units of extraction radius ($0.94c$), outgoing across all three spheres. On the detector track it is the bell that slows from 210 to 171 Hz in Fig. 9(b).

C. The spiral’s carried burst

The spiral’s $(2, 2)$ burst peaks at $|r\Psi_4| = 2.9 \times 10^{-2}$ and folds onto one curve across $R = 20\text{--}44$ at $v/c = 0.95/1.00/1.00$. The burst is fill-free by causality: the interior freeze that carries the late record is armed at $t = 53$ with its skin at $r \geq 1.8$, so the earliest it can influence any extraction sphere is after the burst has passed it—and the head-on’s fill twins bound the device’s imprint at the 0.01%-of-peak level. What the burst is *not* is a black-hole birth cry: the corrected scans find no horizon anywhere on the record (Sec. VII B), so this is the wave of a merger that stalls as a wormhole.

D. The fly-by burst

The loudest signal in the campaign comes from the encounter that makes nothing: the $p = 0.45$ pass emits a single $(2, 2)$ arch of peak $|r\Psi_4| = 4.1 \times 10^{-2}$ with no ringdown—there is no remnant to ring—propagating at $v/c = 1.00$ across every sphere pair. The record is complete to $t = 100$ with no NaN and no horizon at any time; it is gated at $t = 76$, where the mouths’ post-encounter expansion (areal radii $4.2 \rightarrow 33$ by $t = 97$, and $\text{corr}(\log R, \log \|\mathcal{H}\|) = 0.92$ —the swelling, not a wall, drives the late constraint growth) lifts the $R = 20$ sphere off its post-burst trough, while the outer spheres decay monotonically to the record’s end.

E. The vacuum twin

The vacuum control—same d , p and masses, no scalar—does what black holes do: merges and settles onto a 227 Hz ringdown shelf. Against it the wormhole channels are loud: the fly-by’s peak $|r\Psi_4|$ is $3.8\times$ the twin’s and the spiral’s $2.7\times$. The twin also calibrates the

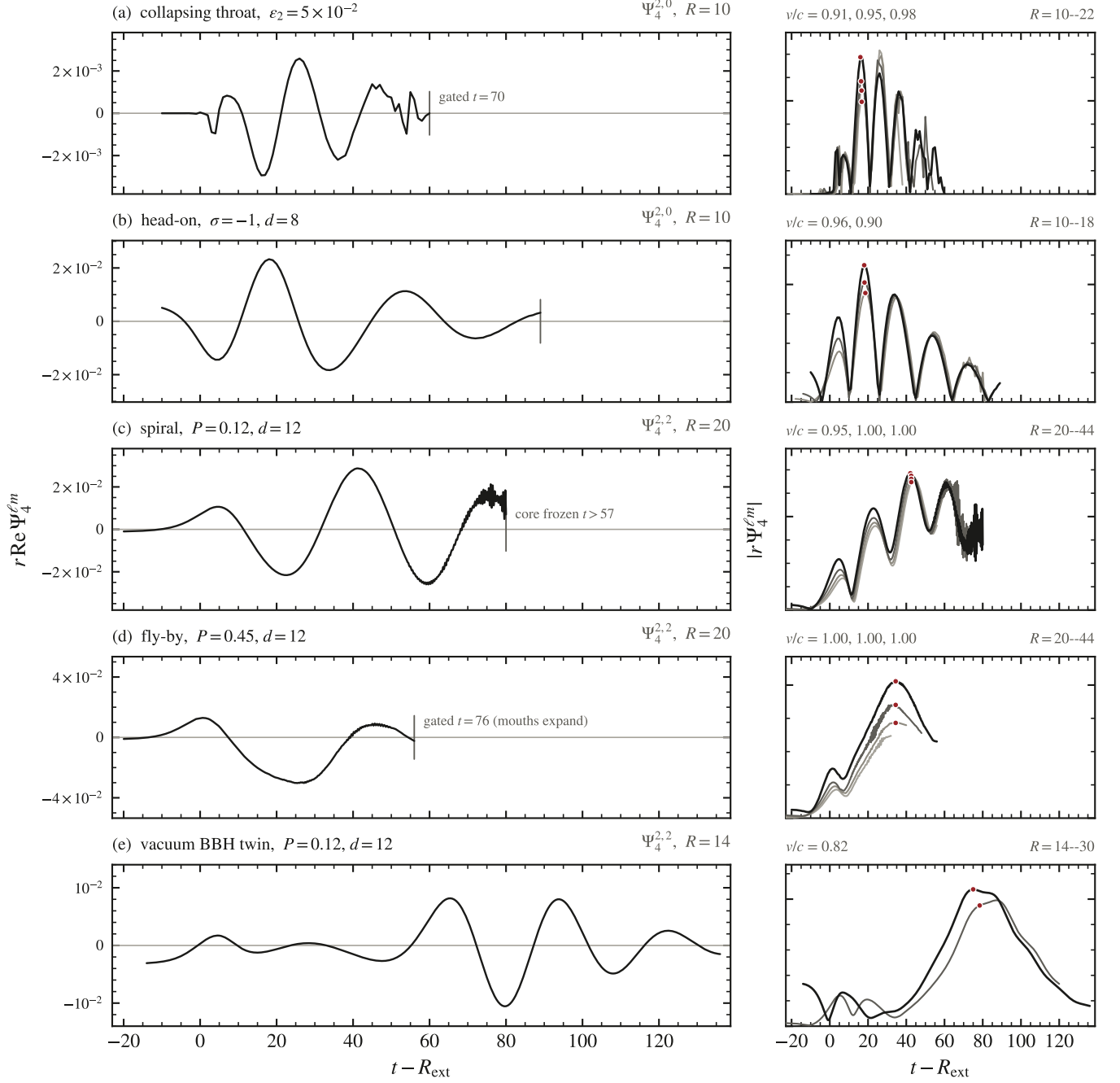


FIG. 8. The waveform gallery: every channel's radiation, one row per scenario, one retarded clock. Left column: $r \text{Re} \Psi_4$ of the dominant mode at the innermost extraction sphere, each row on its own stated scale—the lone throat radiates an order of magnitude below the mergers, and the fly-by is the loudest channel of all. Half-height caps mark record ends: the collapsing throat's stream is junk-gated at $t = 70$, the spiral record ends at $t = 60$ with the run still in flight, and the fly-by—complete to $t = 100$, no horizon ever forming—is gated at $t = 76$, where the mouths' expansion (areal radii growing $\times 7.8$) lifts the $R = 20$ sphere off its post-burst trough. Right column: the envelope $|r \Psi_4|$ at every extraction sphere on its own retarded time (ink to faint, inner to outer): a signal travelling at c and falling as $1/r$ folds onto one curve. Burgundy dots mark the correlated wavefront, and the quoted v/c is the lag of the whole complex waveform between neighbouring spheres over their common retarded window—peak-to-peak timing fails on flat-topped envelopes (the vacuum twin's merger plateau is ~ 15 units wide at both spheres, and its peak times alone would read $v = 0.57c$ where the waveform lag reads $0.82c$). Measured speeds: throat $0.91/0.95/0.98$, head-on $0.96/0.90$, spiral $0.95/1.00/1.00$, fly-by $1.00/1.00/1.00$, vacuum twin 0.82 . Every channel propagates at c within the output cadence: radiation, not a gauge or constraint mode.

wavefront method itself: its merger envelope is a ~ 15 -unit plateau at both spheres, where peak-to-peak timing reads a meaningless $0.57c$ and the full-waveform lag reads $0.82c$ —which is why every speed in this chapter is a correlated lag, never a peak time.

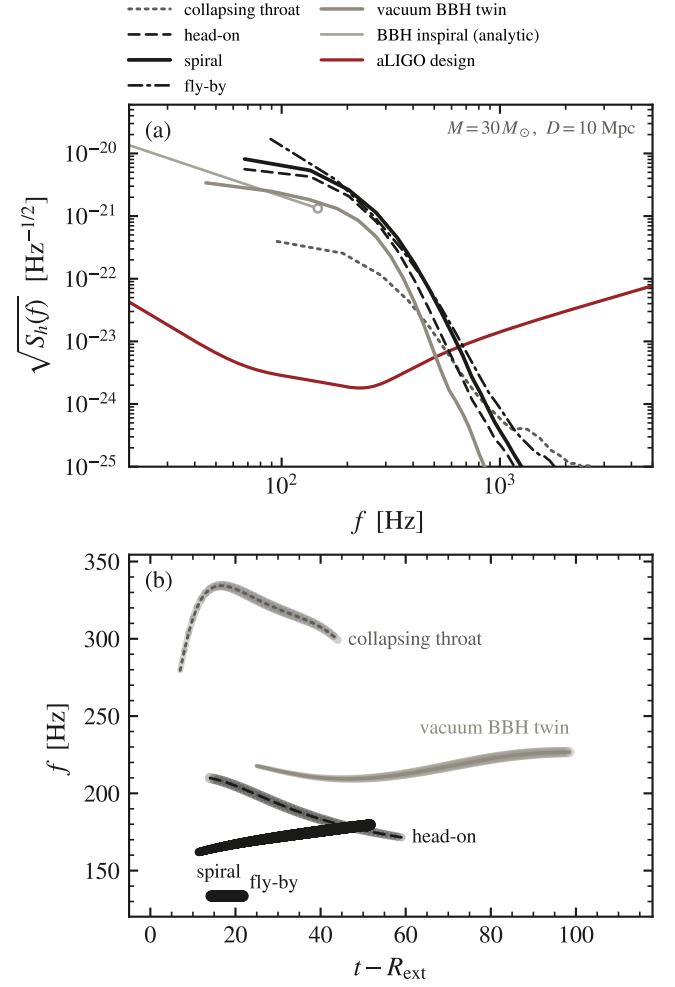


FIG. 9. Every source on a detector, at one shared calibration: total mass $M = 30 M_\odot$, distance $D = 10$ Mpc; strain scales as M/D and frequency as $1/M$. (a) Strain amplitude spectral densities of the innermost-sphere records over the aLIGO design floor (burgundy). Every channel clears the floor around 100 Hz: the fly-by peaks at $1.7 \times 10^{-20} \text{ Hz}^{-1/2}$ (89 Hz); the head-on, spiral and vacuum twin lie within a factor of 2.4 of one another ($3.4\text{--}8.1 \times 10^{-21}$); the lone collapsing throat sits an order below (3.9×10^{-22}). The faint straight line is the analytic Newtonian inspiral of the same equal-mass binary, $|\dot{h}(f)| \propto \mathcal{M}^{5/6} f^{-7/6}$, drawn to its ISCO at 147 Hz through the same normalisation as the numerical record: the ramp an astrophysical binary rides into the band where the simulated signal begins. (b) Peak-frequency tracks of the Morlet wavelet transform of each record (the line's width is the signal's own envelope; tracks are drawn only above 0.2 of each record's peak and outside the cone of influence). The shapes separate the sources: the collapsing throat rings in an arch at 280–334 Hz, the head-on bell slows from 210 to 171 Hz, the vacuum twin climbs through merger onto a 227 Hz ringdown shelf, the spiral (record in flight) enters as a rising arc at 162–180 Hz, and the fly-by burst is a short flat track at 134 Hz.

IX. CONTROLS AND SYSTEMATICS

A. Numerical systematics

Each numerical device carries a twin that measures it. The χ clamp is not load-bearing: twins at 10^{-8} against 10^{-20} re-converge to four digits after touching the floor, and the level-4 pair is byte-identical. The interior fill is not load-bearing for anything outside: head-on fill and no-fill arms agree to 0.01% of the wave peak. The diagnostics are provably non-invasive: two different binaries evolving the same spiral checkpoint produce byte-for-byte identical physics streams over 2384 shared output rows—which also certifies bit-reproducibility on this hardware. The time step is bracketed (dt multiplier 0.05 is a different solution from $t \approx 33$; 0.1 dies at $t = 16$; 0.02 is used throughout), grid independence after the head-on merger is 0.05–0.33% of peak across three restarted arms, and the placement systematic of Sec. VC is subtracted by construction. The Helfer superposition correction is rejected on measurement: better initial data, worse evolution. Matter damping shapes nothing it was hoped to shape.

X. PATTERN MATCHING AGAINST DETECTOR DATA

The simulated waveforms are scale-free: with the throat mass M as the unit, $r\Psi_4$ rescales to detector strain by $h \propto M/D$ and $f \propto 1/M$, so one numerical waveform covers every candidate mass at once. This section confronts the rescaled (2,2) and (2,0) waveforms with the LIGO–Virgo–KAGRA data products on three fronts: matched filtering [26], the ringdown, and the unmodelled searches.

A. Fitting factor against the binary-black-hole bank

B. Ringdown discriminators

C. Burst searches and the background

XI. DISCUSSION

XII. REPRODUCIBILITY

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